Suman Chowdhury

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RESEARCH FOCUS

Scene Graphs • Vision-Language Models • Event Recognition • Causal Reasoning • Autonomous Vehicles

EDUCATION

INTEGRATED M.TECH–PHD IN ENGINEERING SCIENCES
Aug 2021 – Present

Academy of Scientific and Innovative Research (AcSIR)
- Advisor: Dr. Rajesh P. Barnwal, CSIR-CMERI
- Cumulative GPA: 8.57/10.0
- M.Tech Thesis: Multi-Label Classification for Road Scene Understanding

B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING
Aug 2015 – Jun 2019

Jalpaiguri Government Engineering College, WB
- Cumulative GPA: 7.99/10.0
- Thesis: Distributed Storage System using Optimal Exact-Regenerating Codes

RESEARCH EXPERIENCE

CSIR SENIOR RESEARCH FELLOW (GATE)
Nov 2023 – Present

AI & IoT Lab, IT Group, CSIR–CMERI, Durgapur, India
- Leading research on traffic scene understanding, event recognition, and causal reasoning for autonomous vehicles.
- Current focus: integrating Vision-Language Models (VLMs) and Scene Graphs for interpretable AV decision-making.
- Completed first PhD objective on multi-label activity recognition in traffic scenes (under review at IEEE TCSVT).

CSIR JUNIOR RESEARCH FELLOW (GATE)
Nov 2021 – Nov 2023

AI & IoT Lab, IT Group, CSIR–CMERI, Durgapur, India
- Developed multi-label learning framework for road scene understanding.
- Initiated survey on high-level scene understanding in traffic environments (under review at ACM CSUR).

SUMMER RESEARCH INTERN (UG)
Jun 2018

Dept. of EECE, IIT Kharagpur, India
- Built a practical distributed storage system prototype using exact-regenerating codes.
- Implemented data storage/retrieval and node failure recovery.
- Advisor: Prof. Sant Sharan Pathak

PUBLICATIONS

ACCEPTED MANUSCRIPTS

Sahu, Bidya, Suman Chowdhury, and Rajesh P. Barnwal. “Near Real Time Detection of APT Attack in Industrial IoT Environment.”
Conference: ICDCN 2026

Chowdhury, Suman, Farook, Yahya, and Rajesh P. Barnwal. “Real-Time Vision-based Scene Classification for Adaptive Behavior in Autonomous Mobile Robots.”
Conference: ACM CODS-COMAD December-2024

Chakraborty, Koustav, Suman Chowdhury, and Rajesh P. Barnwal. “Study on the Efficacy of GPT Double Heads Model for Question-Answering in Bengali.”
Conference: IEEE Calcon 2024

MANUSCRIPTS UNDER REVIEW

Chowdhury, Suman, Prerana Mukerjee, and Rajesh P. Barnwal. “GraSS: Graph-Guided Semantic Slot Attention for Multi-Label Video Activity Recognition in Autonomous Agents.”
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

Chowdhury, Suman, Prerana Mukerjee, and Rajesh P. Barnwal. "Vision-Based High-Level Road Scene Understanding for Autonomous Vehicles: State of the Art, Challenges, and Future Directions."
ACM Computing Surveys (ACM CSUR)

Chowdhury, Suman, and Rajesh P. Barnwal. "Empirical Study of Transformer and Graph-Based Architectures for Class-Imbalanced Road Scene Recognition." Available at SSRN 4718551.

MENTORING EXPERIENCE

Supervised multiple B.Tech interns (2023–25) on projects in deep learning, scene understanding, and vision-based AVs.
Outcomes include 10 completed projects and 3 accepted conference publications.

ACHIEVEMENTS

Awarded Travel Grant – ACM CODS-COMAD 2024 Conference — December 2024
Selected for a travel grant to attend the ACM CODS-COMAD conference.
- Link: cods-comad.in/dec-2024/travel-grants.php

ACM India Anveshan Setu Fellowship 2024–25 — February 2024
Selected among 46 CS/IT PhD scholars across India for a prestigious mentorship program with leading researchers. As part of the fellowship, visited Prof. Prerana Mukherjee at Jawaharlal Nehru University in March 2025.
- Link: acm.org/articles/acm-india-bulletins/2024/announcing-acm-india-anveshan-setu-fellows

CSIR Junior Research Fellowship (GATE) — November 2021 – November 2026
Awarded to a maximum of five scholars per CSIR national laboratory to pursue the Integrated M.Tech–PhD program at AcSIR through GATE qualification.

Full Fee Scholarship (FFS) — August 2015 – June 2019
Granted full tuition fee waiver for the 4-year B.Tech program at Jalpaiguri Government Engineering College by the Government of West Bengal, based on academic merit and financial need.

JEE Advanced 2015 – Common Merit List — June 2015
Secured AIR 16,887 (General category) out of 1.2 million candidates. Also qualified the Architecture Aptitude Test (AAT) 2015, organized by IIT Bombay.

SKILLS

PROGRAMMING
Experienced: Python | GNU Octave • Familiar: C | C++ | MATLAB | JAVA

FRAMEWORKS & LIBRARIES
PyTorch • Tensorflow • Numpy • Pandas • Scikit-learn • Matplotlib

LANGUAGES
Native: Bengali • Fluent: English | Hindi

VOLUNTEERING

Secretary — JULY 2024 - PRESENT
AcSIR Science Club
CSIR-Central Mechanical Engineering Research Institute

Teacher — JANUARY 2016 - JUNE 2019
Jyoti- A free Night School for underprivileged students
Jalpaiguri Government Engineering College